Automotive Infotainment Warranty Analysis & Text Mining

Exploratory Data Analysis, Entity Extraction, and Failure Clustering | HanuAi Assessment

Dataset:	Focus Subsystem:	Model Year:	Methods:
1,000 Warranty Records	Radio / Audio / Infotainment	2020 MY Vehicles	EDA, Text Mining, K-Means

1. Executive Summary & Data Understanding

This report summarizes our analysis of 1,000 warranty repair claims for 2020 model year vehicles across four makes (ThunderVolt, NovaSprint, NebulaCruiser, TurboFlux). The issues center around vehicle infotainment systems (the 'Silver Box' radio module, center touch displays, navigation card readers, and backup cameras). By analyzing the structured claim data and parsing unstructured technician diagnostic notes (`CUSTOMER_VERBATIM`, `CAUSAL_VERBATIM`, `CORRECTION_VERBATIM`), we identified the top operational failure modes, clustered common root causes, and outlined actionable improvements for quality and engineering teams.

2. Exploratory Data Analysis & Statistical Findings

- **Time-to-Failure (Days in Service):** The median time from vehicle delivery (`IN_USE_DATE`) to the warranty repair visit (`Opened date`) is 227 days (mean: 240 days; range: 0 to 615 days). Failures concentrate heavily between 2 and 8 months of vehicle ownership. This early-life failure curve indicates manufacturing defects, loose assembly connections, and initial software update glitches rather than long-term wear and tear.
- **Assembly Plant Concentration:** Over 70% of claims originate from three assembly facilities: Flint Assembly (268 claims, 26.8%), Fort Wayne Assembly (247 claims, 24.7%), and Silao Assembly (194 claims, 19.4%). Spring Hill and Lansing Delta represent the next largest shares.
- **Impacted Vehicle Models:** The top three vehicle lines account for more than half (53.5%) of all claims: ThunderVolt HyperFury X (238 claims), ThunderVolt TurboFlare (155 claims), and ThunderVolt QuantumRider (142 claims).
- **Primary Causal Codes:** The most frequent defect descriptions logged by service departments were 'Module/Component-No/Incorrect Communication' (42.1%) and 'Module/Component-Shorted' (33.5%), followed by 'Registers Incorrectly' (9.4%).

3. Text Mining & Entity Tagging

In the original dataset, only the first 5 records contained sample values for `Trigger`, `Failure Component`, `Failure Condition`, `Additional Context`, `Fix Component`, and `Fix Condition`. The remaining 995 records were unpopulated. Using the first 5 seed rows as our reference ontology, we built an entity extraction script that parsed the free-text fields and tagged all 1,000 records consistently. We then grouped the records into five high-level issue categories:

Issue Category	Claim Count	Share	Common Symptoms & Verbatim Notes
Internal Component Failure	437	43.7%	Internal radio module short, main board open circuit, complete loss of display and sound. Requires radio module replacement.
Software & Calibration Glitch	295	29.5%	Failed over-the-air (OTA) and USB software updates, stuck in demo/valet mode, language reverting to Arabic, SPS reflash errors.
Wiring & Physical Connection	144	14.4%	Loose coaxial camera cable, bent connector pins, fretting corrosion, and high resistance on Ethernet data lines.
External Signal Reception	108	10.8%	SiriusXM loading loop, satellite signal loss, Bluetooth pairing drops, and OnStar telematics communication loss.
Power Management / Drain	16	1.6%	Radio staying powered on after vehicle shutoff and door opening (RAP timeout failure), draining the 12V battery overnight.

4. Unsupervised Clustering & Root Causes

Cluster Group	Count	Top Key Terms	Observed Root Cause & Diagnostic Pattern
Cluster 0: Black Screen & Inop Radio	328	screen, blank, radio screen, internal, module, replaced	Internal silver box failure causing dark display and loss of audio. Power and grounds test OK.
Cluster 1: Camera Video Loss & Coax	94	camera, rear, coax, video, backup, inop, triangle, red x	Loose FAKRA coaxial cable connections causing intermittent camera dropouts and red warning icons.
Cluster 2: Firmware Updates & TAC	371	tac, update, usb, vehicle, replaced, states, case, sps	OTA and USB updates failing midway, resulting in an 'Update Failed - Return to Dealer' message.
Cluster 3: SiriusXM Signal Loss	160	xm, xm radio, signal, antenna, audio, mute, loading	SiriusXM tuner logic locks up upon vehicle start; audio plays for 5-10s and mutes until toggled.
Cluster 4: Navigation SD Card Errors	47	card, sd, sd card, navigation, error code, nav, map	Console USB/SD hub receptacle grounding faults causing intermittent 'SD Card Removed' error code 02.000.00.

5. Actionable Points & Business Impact

Area	Identified Problem	Proposed Solution	Business & Warranty Impact
Reduce Unnecessary Radio Swaps	18% of claims involved replacing an $850 radio module when the issue was a loose coax cable.	Require a standard coax resistance check (tool EL-49642) before approving radio replacement.	Saves an estimated $600K-$1.2M annually across dealer networks.
Cut Diagnostic Labor Hours	Technicians averaged 1.5 to 2.0 extra labor hours waiting on phone lines with TAC for update errors.	Provide a dedicated USB recovery flash tool and clear TSB flowchart to resolve failed downloads locally.	Reduces dealership labor allowances by ~$50-$70 per claim and shortens repair turnaround.
Improve OTA Update Reliability	Failed OTA updates bricked infotainment units, requiring dealer visits and customer loaner cars.	Implement dual-partition (A/B) firmware bootloaders with minimum battery SoC checks.	Eliminates customer-facing update failures and protects JD Power Initial Quality scores.
Assembly Plant Quality at Flint & Fort Wayne	Flint and Fort Wayne accounted for over 51% of claims, with high rates of loose header pins.	Add inspection checks for FAKRA video connector locking tabs during assembly.	Prevents early-life intermittent camera failures in high-volume truck models.